

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

1. Section A (MCQs): Attempt ALL 12 questions. Each carries 1 mark. Time: 15 minutes.
2. Section B (Short Questions): Attempt any 11 of the 14 questions given. Each carries 3 marks.
3. Section C (Long Questions): Attempt any 4 of the 5 questions given. Each carries 5 marks.
4. Use black or blue ink only. Pencil is not allowed except for diagrams and number lines.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

Encircle / bubble the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The multiplicative inverse of 0 is:	0	1	-1	undefined	(A)(B)(C)(D)
2	In the exponential form $5^{2/3}$, the denominator of the exponent (3) represents:	the base	the index of the radical	the radicand	the coefficient	(A)(B)(C)(D)
3	The mantissa of a logarithm is always:	negative	zero or positive, never negative	an integer	equal to the characteristic	(A)(B)(C)(D)
4	Which of the following logarithms is undefined?	$\log_3 9$	$\log_{10} 1000$	$\log_2(-4)$	$\log_5 25$	(A)(B)(C)(D)
5	In the number 0.00532, the reference position lies:	between 5 and 3	before the decimal point	after the last digit	at the decimal point	(A)(B)(C)(D)
6	The HCF of two polynomials P and Q remains unchanged if both are:	multiplied by the same nonzero constant	multiplied by different variables	divided by different constants	added together	(A)(B)(C)(D)
7	The factors of $x^2 - 11x + 30$ are:	$(x-5)(x-6)$	$(x-3)(x-10)$	$(x+5)(x+6)$	$(x-2)(x-15)$	(A)(B)(C)(D)
8	The square root of $49a^2 - 84ab + 36b^2$ is:	$\pm(7a-6b)$	$\pm(7a+6b)$	$\pm(49a-36b)$	$7a-6b$ only	(A)(B)(C)(D)
9	The solution set of $ x - 4 = 7$ is:	{11, -3}	{3, -11}	{11, 3}	{-3, -11}	(A)(B)(C)(D)
10	If $3 < 5$ and $c = -2$, which statement is true?	$3(-2) < 5(-2)$	$3(-2) > 5(-2)$	$3(-2) = 5(-2)$	cannot be determined	(A)(B)(C)(D)
11	The point (3, -5) lies in which quadrant?	First	Second	Third	Fourth	(A)(B)(C)(D)
12	If P is the midpoint of line segment RS and PS = 6 units, then RS =	3	6	12	18	(A)(B)(C)(D)

SECTION B — Short Questions (Attempt any 11 of $14 \times 3 = 33$ Marks)

Attempt any 11 of the following 14 questions. Show all working.

Part	Question	Marks
2(i)	Momina walks at 3.5 km/h. If she walks continuously for 12 hours, how far does she walk? Express your answer in scientific notation.	3
2(ii)	The world's lowest recorded temperature was -89.2°C (Vostok, Antarctica, 1983). Convert this temperature into Fahrenheit and Kelvin scales, using $F = (9/5)C + 32$ and $K = C + 273$.	3

2(iii)	Identify the property of equality or inequality of real numbers that justifies each statement: (a) If $x + 2 = y$ and $y = 2x - 3$, then $x + 2 = 2x - 3$. (b) If $3 < 4$, then $-3 > -4$.	3
2(iv)	Find the antilog of: (a) 2.4324 (b) 1.5890	3
2(v)	Check whether $\log_2(7 - x)$ is defined for: (a) $x = 0$ (b) $x = 1$ (c) $x = 6$ (d) $x = 7$	3
2(vi)	Verify that $\log_c a \cdot \log_a b = \log_c b$, using $a = 4$, $b = 64$ and $c = 2$.	3
2(vii)	Factorize completely: (a) $27 + 512x^3$ (b) $a^6 - 64$	3
2(viii)	Find the HCF of $12x^2 + x - 1$ and $15x^2 + 8x + 1$ by factorization.	3
2(ix)	Find the square root of $25a^4 - 30a^3 + 9a^2$ by the factorization method.	3
2(x)	Two times a number decreased by 5 is greater than or equal to the number increased by 8. Find the possible values of the number.	3
2(xi)	Check whether the given value of the variable satisfies the inequality: (a) $5y - 12 > 0$; $y = 3$ (b) $5 - 2x > -4x + 5$; $x = 4$	3
2(xii)	Solve the absolute value equation: $ 5x - 3 = x + 7 $	3
2(xiii)	Find the midpoint of the line segment joining (1.4, -1.5) and (2.6, 3.5).	3
2(xiv)	Three vertices of rectangle ABCD are A(1, 2), B(7, 2) and C(7, 8). Using the property that adjacent sides of a rectangle are parallel to the axes, find the coordinates of vertex D.	3

SECTION C — Long Questions (Attempt any 4 of $5 \times 5 = 20$ Marks)

Attempt any 4 of the following 5 questions. Most carry internal parts — show all working.

Part	Question	Marks
3(i)	(a) Verify the Trichotomy Property of real numbers for $a = -2\sqrt{3}$ and $b = -5$. [2] (b) A company was penalized for a compliance violation. It has three partners — Aatika, Bilal and Faryal — who invested in the ratio 1 : 2 : 3. If the total penalty of Rs. 600,000 is to be shared in this ratio, find the amount paid by each partner. [3]	5
3(ii)	Given $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 5 = 0.6990$, $\log 7 = 0.8451$: (a) Evaluate $\log(28/45)$ using the laws of logarithms, showing each step. [3] (b) Find the number of digits in 7^{40} . [2]	5
3(iii)	The volume of a cubical oil tank in an oil refinery is expressed as $(125x^3 - 150x^2 + 60x - 8)$ m^3 . Find: (a) the expression for the height (side) of the tank. [2] (b) the expression for the total surface area of the tank. [1] (c) the expression for the cost of painting it from outside at Rs. 32/ m^2 . [2]	5
3(iv)	(a) The sum of five times a number x and 10 is less than -35 or greater than -5 . Find the set of real numbers that x can represent. [3] (b) Solve the equation $ 2y - 5 = 9$ and represent the solution set on a number line. [2]	5
3(v)	Quadrilateral ABCD has vertices A(1, 0), B(5, 2), C(2, 6), D(-3, 2). E, F, G, H are the midpoints of AB, BC, CD, DA respectively. (a) Find the coordinates of E, F, G, H. [2] (b) Find the lengths of EF, FG, GH, HE. [2] (c) What type of quadrilateral is EFGH? Justify briefly. [1]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry)

— End of Question Paper —